EXPERIMENT NO. 9

Data Storytelling and Business Insight Generation

using Interactive Visualizations

NAME : Krishna Sharma

UID : CU240250899

COURSE : BTECH CSE

SECTION : B

ROLL NO. : 32

AIM

To create a compelling data-driven story by analyzing a real-world dataset, developing meaningful visualizations, and presenting actionable business insights.

STEP 1: IMPORT REQUIRED LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import tkinter as tk

from tkinter import filedialog
```

STEP 2: SELECT DATASET FROM COMPUTER

```
from google.colab import files
import io

print("Please upload your dataset file (e.g., CSV).")
uploaded = files.upload()

if not uploaded:
    raise ValueError("No file selected.")

# Assuming only one file is uploaded for simplicity
filename = next(iter(uploaded))

try:
    # Read the CSV file into a pandas DataFrame
    df = pd.read_csv(io.BytesIO(uploaded[filename]))
    print(f"Successfully loaded {filename} into a DataFrame.")
except Exception as e:
    raise ValueError(f"Unable to read the selected file as a CSV: {e}")
```

```
Please upload your dataset file (e.g., CSV).
Choose Files Global Superstore.csv
Global Superstore.csv(text/csv) - 12444106bytes,lastmodified:5/23/2026 - 100% done
Saving Global Superstore.csv to Global Superstore.csv
Successfully loaded Global Superstore.csv into a DataFrame.
```

STEP 3: BASIC DATASET ANALYSIS

```
print("Dataset Shape:", df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nFirst Five Rows:")
print(df.head())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
2  Office  Supplies  Los  Angeles United  States  CS 121304 AP-     Chad Sievert
3  Office  Supplies  Los  Angeles United  States  121304 AP-     Chad Sievert
4  Office Supplies Los Angeles   United States   109154       Arthur Prichep

     Discount Market ji_lu-shu Order Date        Order ID ... Sales   Segment  \
0       0.0     US          1     00:00.0 CA-2011-130813   ...   19  Consumer
1       0.0     US          1     00:00.0 CA-2011-148614   ...   19  Consumer
2       0.0     US          1     00:00.0 CA-2011-118962   ...   21  Consumer
3       0.0     US          1     00:00.0 CA-2011-118962   ...  111  Consumer
4       0.0     US          1     00:00.0 CA-2011-146969 ...      6  Consumer

  Ship Date       Ship Mode  Shipping Cost       State  Sub-Category  Year  \
0   00:00.0    Second Class           4.37  California         Paper  2011
1   00:00.0  Standard Class           0.94  California         Paper  2011
2   00:00.0  Standard Class           1.81  California         Paper  2011
3   00:00.0  Standard Class           4.59  California         Paper  2011
4   00:00.0  Standard Class           1.32  California         Paper  2011

         Market2 weeknum
0  North   America      2
1  North   America      4
2  North   America     32
3  North   America     32
4  North America      40

[5 rows x 27 columns]

Missing Values:
Category          0
City              0
Country           0
Customer     ID   0
Customer   Name   0
Discount Market   0
ji_lu-shu Order   0
Date Order ID     0
Order  Priority   0
Product      ID   0
Product    Name   0
Profit Quantity   0
Region            0
Row ID            0
Sales             0
Segment           0
Ship Date         0
Ship       Mode   0
Shipping   Cost   0
State      Sub-   0
Category Year     0
Market2 weeknum   0
dtype: int64      0
                  0
                  0
                  0
                  0



Duplicate Rows: 0
```

STEP 4: CLEAN THE DATA

```
df = df.drop_duplicates()

# Convert column names into strings
df.columns = df.columns.astype(str).str.strip()

# Convert date columns if available
for column in df.columns:
    if "date" in column.lower():
        try:
            df[column] = pd.to_datetime(df[column])
        except:
            pass
```

STEP 5: IDENTIFY IMPORTANT BUSINESS COLUMNS

```
sales_col = None
profit_col = None
category_col = None
region_col = None

for column in df.columns:

    name = column.lower()

    if sales_col is None and "sales" in name:
        sales_col = column

    if profit_col is None and "profit" in name:
        profit_col = column

    if category_col is None and "category" in name:
        category_col = column
```

```
    if region_col is None and "region" in name:
        region_col = column
```

STEP 7: KEY PERFORMANCE INDICATORS (KPIs)

```
if sales_col is not None:

    df[sales_col] = pd.to_numeric(
        df[sales_col],
        errors="coerce"
    )

    total_sales = df[sales_col].sum()
    average_sales = df[sales_col].mean()

    print("\n========== KEY PERFORMANCE INDICATORS ==========")
    print("Total Sales:", round(total_sales, 2))
    print("Average Sales:", round(average_sales, 2))

if profit_col is not None:

    df[profit_col] = pd.to_numeric(
        df[profit_col],
        errors="coerce"
    )

    total_profit = df[profit_col].sum()
    average_profit = df[profit_col].mean()

    print("Total Profit:", round(total_profit, 2))
    print("Average Profit:", round(average_profit, 2))
```

```
========== KEY PERFORMANCE INDICATORS ==========
Total Sales: 12642905
Average Sales: 246.5
Total Profit: 1467457.29
Average Profit: 28.61
```

STEP 8: SALES BY CATEGORY

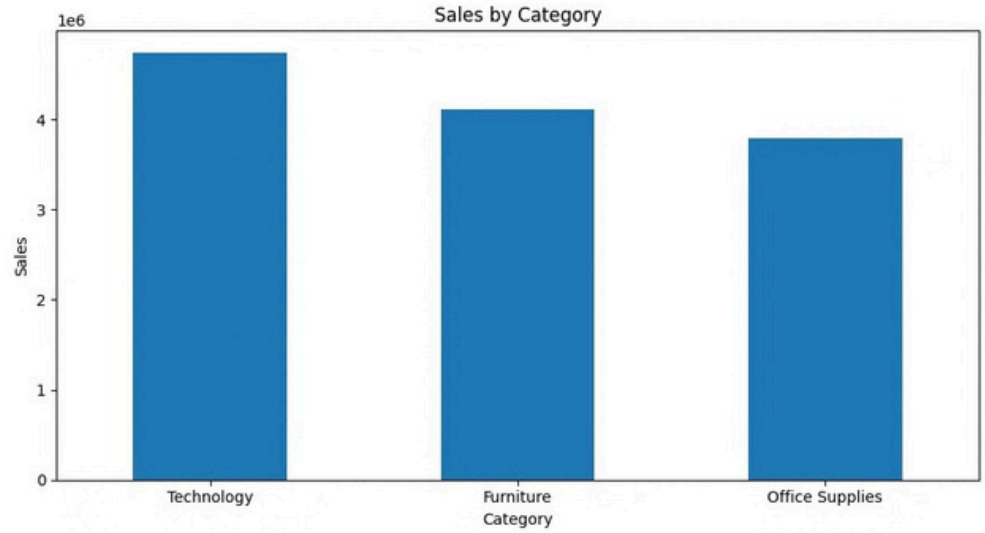
```
if sales_col is not None and category_col is not None:

    category_sales = (
        df.groupby(category_col)[sales_col]
        .sum()
        .sort_values(ascending=False)
    )

    plt.figure(figsize=(9, 5))

    category_sales.plot(kind="bar")

    plt.title("Sales by Category")
    plt.xlabel("Category")
    plt.ylabel("Sales")
    plt.xticks(rotation=0)
    plt.tight_layout()
    plt.show()
else:
    print("Could not generate 'Sales by Category' plot. No column containing 'category' was found in the dataset.")
```



STEP 9: PROFIT BY CATEGORY

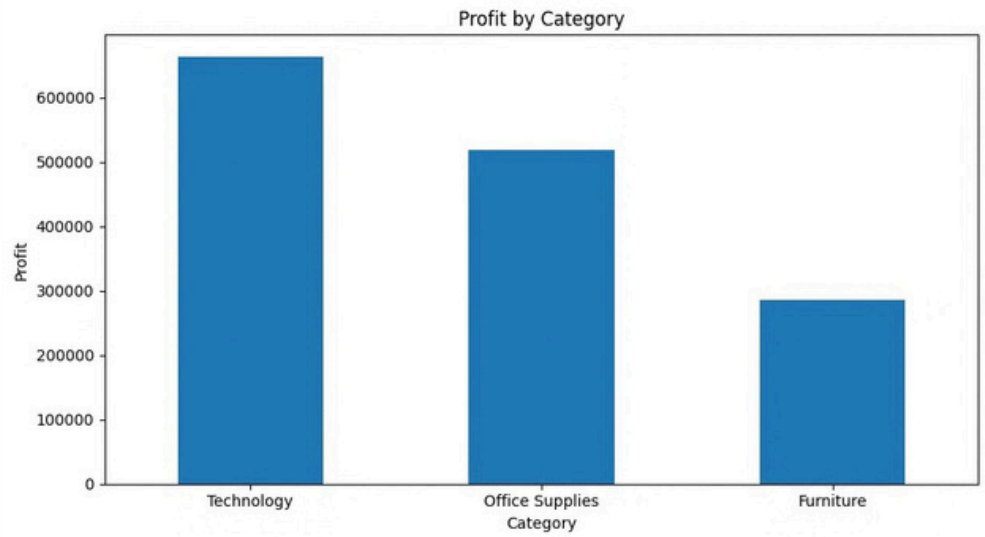
```
if profit_col is not None and category_col is not None:

    category_profit = (
        df.groupby(category_col)[profit_col]
        .sum()
        .sort_values(ascending=False)
    )

    plt.figure(figsize=(9, 5))

    category_profit.plot(kind="bar")

    plt.title("Profit by Category")
    plt.xlabel("Category")
    plt.ylabel("Profit")
    plt.xticks(rotation=0)
    plt.tight_layout()
    plt.show()
```



STEP 10: SALES BY REGION

```
if sales_col is not None and region_col is not None:

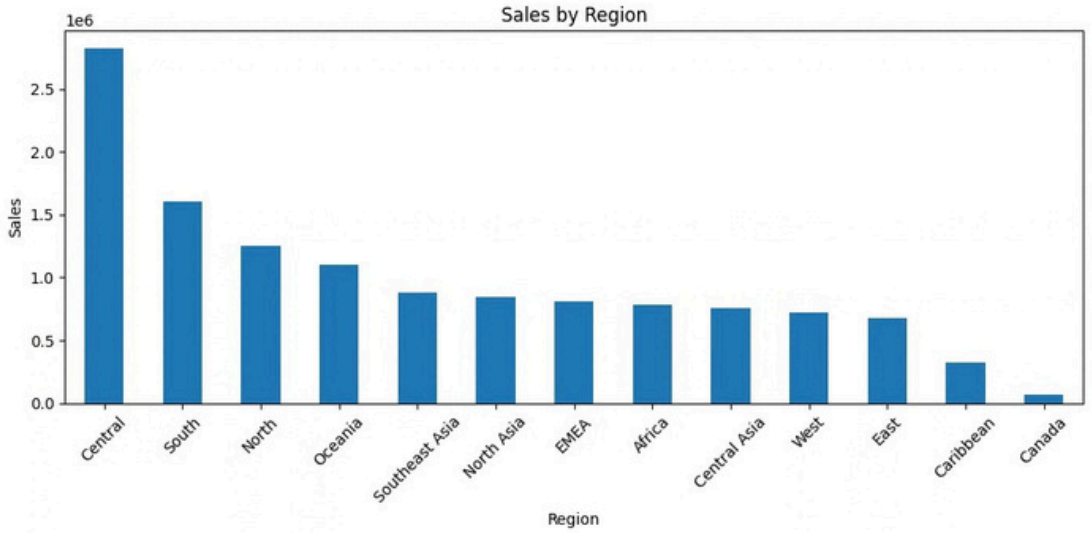
    region_sales = (
        df.groupby(region_col)[sales_col]
        .sum()
        .sort_values(ascending=False)
    )

    plt.figure(figsize=(10, 5))

    region_sales.plot(kind="bar")

    plt.title("Sales by Region")
```

```
    plt.xlabel("Region")
    plt.ylabel("Sales")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```



STEP 11: PROFIT BY REGION

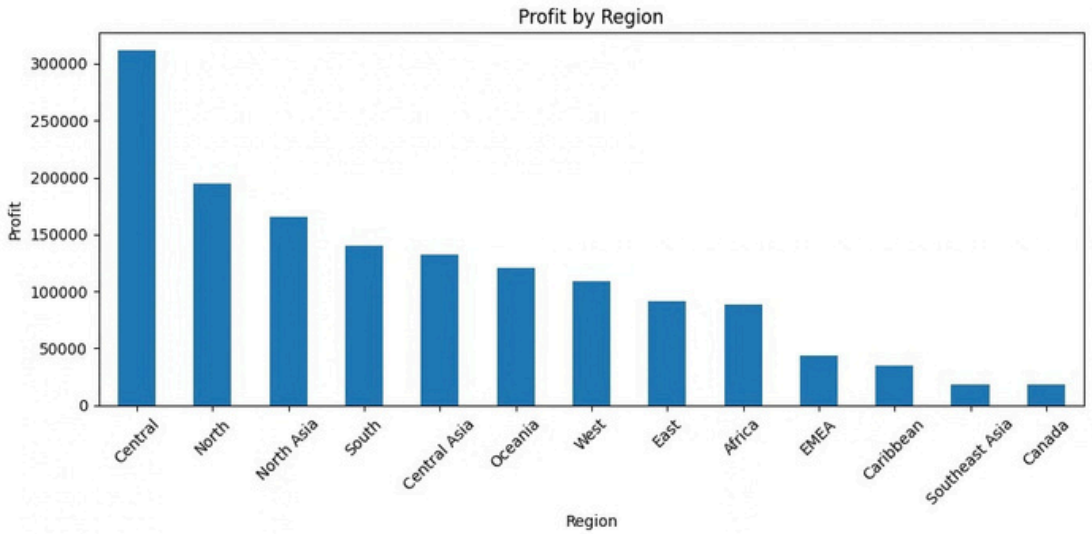
```
if profit_col is not None and region_col is not None:

    region_profit = (
        df.groupby(region_col)[profit_col]
        .sum()
        .sort_values(ascending=False)
    )

    plt.figure(figsize=(10, 5))

    region_profit.plot(kind="bar")

    plt.title("Profit by Region")
    plt.xlabel("Region")
    plt.ylabel("Profit")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```



STEP 12: CORRELATION HEATMAP

```
numeric_df = df.select_dtypes(include=np.number)

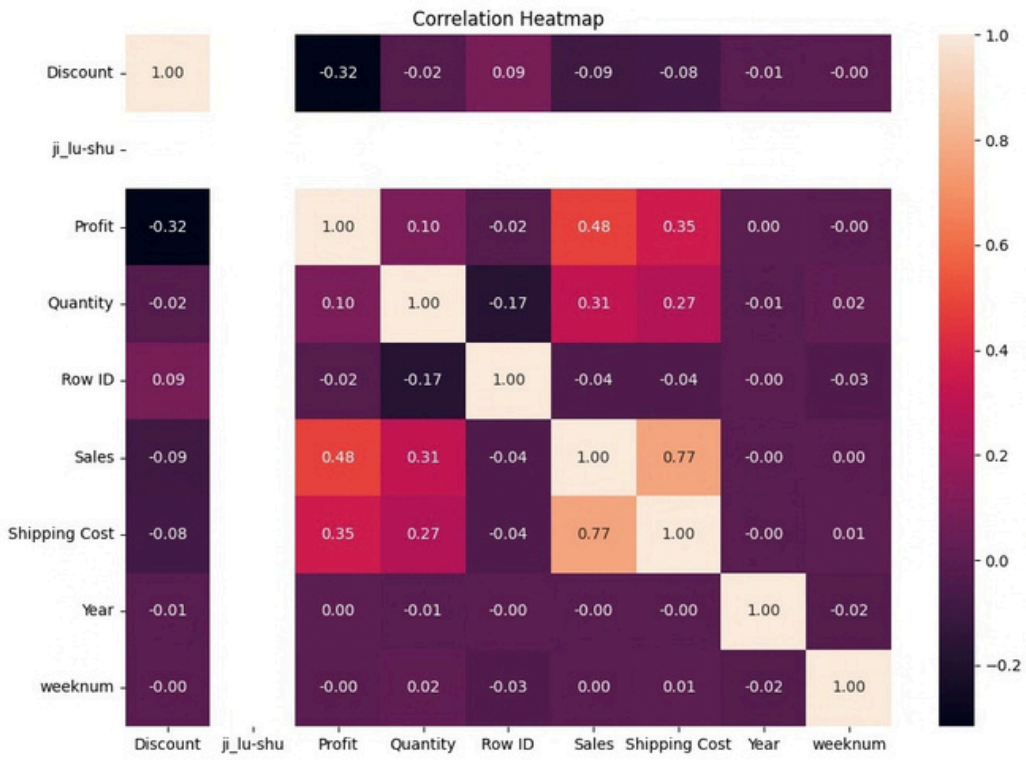
if numeric_df.shape[1] >= 2:

    plt.figure(figsize=(10, 7))

    sns.heatmap(
```

```
        numeric_df.corr(),
        annot=True,
        fmt=".2f"
    )

    plt.title("Correlation Heatmap")
    plt.tight_layout()
    plt.show()
```



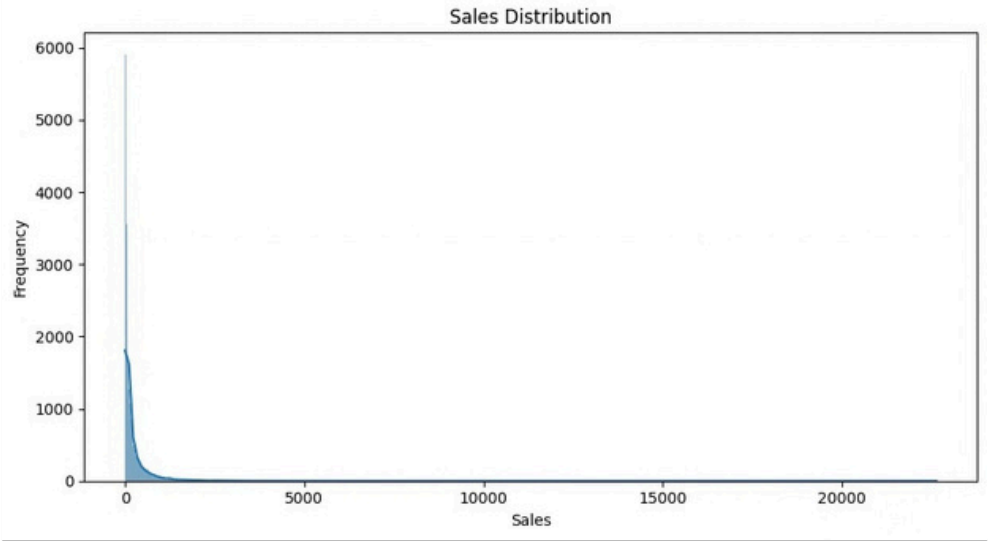
STEP 13: SALES DISTRIBUTION

```
if sales_col is not None:

    plt.figure(figsize=(9, 5))

    sns.histplot(
        df[sales_col].dropna(),
        kde=True
    )

    plt.title("Sales Distribution")
    plt.xlabel("Sales")
    plt.ylabel("Frequency")
    plt.tight_layout()
    plt.show()
```



STEP 14: SALES VS PROFIT

```
if sales_col is not None and profit_col is not None:

    plt.figure(figsize=(9, 6))

    sns.scatterplot(
    data=df,
    x=sales_col,
    y=profit_col
    )

    plt.title("Sales vs Profit")
    plt.xlabel("Sales")
    plt.ylabel("Profit")
    plt.tight_layout()
    plt.show()
```



STEP 15: MONTHLY SALES TREND

```
date_col = None

for column in df.columns:
    if pd.api.types.is_datetime64_any_dtype(df[column]):
        date_col = column
        break

if date_col is not None and sales_col is not None:

    monthly_sales = (
        df.set_index(date_col)
        .resample("ME")[sales_col]
        .sum()
    )

    plt.figure(figsize=(12, 5))

    monthly_sales.plot()

    plt.title("Monthly Sales Trend")
    plt.xlabel("Month")
    plt.ylabel("Sales")
    plt.grid()
    plt.tight_layout()
    plt.show()
```

STEP 16: TOP BUSINESS RECORDS

```
if sales_col is not None:

    top_records = df.nlargest(10, sales_col)

    print("\n========== TOP 10 SALES RECORDS ==========")
    print(top_records[[sales_col]].to_string(index=False))
```

```
========== TOP 10 SALES RECORDS ==========
Sales
 22638
 17500
 14000
 11200
 10500
9893
9450
9100
8750
8400
```

```
print("\n========== BUSINESS INSIGHTS ==========")

if sales_col is not None:
    print("1. Sales performance was analyzed using overall sales and category-wise sales.")

if profit_col is not None:
    print("2. Profitability was analyzed to identify financially strong areas.")

if category_col is not None:
    print("3. Category-level performance was compared to identify important product categories.")

if region_col is not None:
    print("4. Regional performance was analyzed to identify high and low performing regions.")

print("5. Correlation analysis was used to understand relationships between numerical variables.")
```

```
========== BUSINESS INSIGHTS ==========
1. Sales performance was analyzed using overall sales and category-wise sales.
2. Profitability was analyzed to identify financially strong areas.
3. Category-level performance was compared to identify important product categories.
4. Regional performance was analyzed to identify high and low performing regions.
5. Correlation analysis was used to understand relationships between numerical variables.
```

STEP 18: BUSINESS RECOMMENDATIONS

```
print("\n========== BUSINESS RECOMMENDATIONS ==========")

print("1. Focus marketing efforts on high-performing categories.")
print("2. Investigate low-performing categories and identify reasons for weak performance.")
print("3. Strengthen strategies in regions showing strong sales and profitability.")
print("4. Analyze low-profit or loss-making transactions for cost optimization.")
print("5. Use sales trends for inventory and resource planning.")
```

```
========== BUSINESS RECOMMENDATIONS ==========
1. Focus marketing efforts on high-performing categories.
2. Investigate low-performing categories and identify reasons for weak performance.
3. Strengthen strategies in regions showing strong sales and profitability.
4. Analyze low-profit or loss-making transactions for cost optimization.
5. Use sales trends for inventory and resource planning.
```

QUESTIONS AND ANSWERS

Q1. What is data storytelling? How is it different from data visualization?

Answer: Data storytelling combines data analysis, visualization, and narrative to communicate meaningful insights. Data visualization mainly presents information through charts and graphs, while data storytelling also explains the meaning and importance of those findings.

Q2. Why is storytelling important in business analytics and decision-making?

Answer: Storytelling helps stakeholders understand analytical findings easily. It connects data with business problems and helps communicate insights that can support business decisions.

Q3. What are the essential components of an effective data story?

Answer: The main components are:

Business problem or objective

Relevant data

Analysis

Meaningful visualizations

Key insights

Recommendations

Clear narrative

Q4. How do Key Performance Indicators (KPIs) enhance business reporting?

Answer: KPIs provide measurable indicators of business performance. They allow organizations to monitor important metrics such as sales, profit, revenue, growth, and customer performance.

Q5. Why should visualizations be arranged in a logical sequence?

Answer: A logical sequence helps the audience understand the story step by step. It normally starts with the problem, continues with analysis and findings, and ends with insights and recommendations.

Q6. What factors should be considered while selecting visualizations for a business presentation?

Answer: The type of data, purpose of analysis, audience, readability, comparison requirements, number of variables, and amount of information should be considered while selecting a visualization.

Q7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Answer: Dashboards provide an interactive view of important business metrics. Storytelling explains the meaning of those metrics and connects them with business objectives. Together they make analytical information easier to understand and use.

Q8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Answer: Common challenges include technical terminology, complex charts, excessive information, unclear explanations, and difficulty connecting analytical results with business objectives.

Q9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Answer:

Retail companies can use sales and customer data to identify high-performing products and adjust inventory and marketing.

Public health organizations can use disease and vaccination data to communicate trends and support public health planning.

Q10. How can effective data storytelling improve strategic planning and organizational performance?